

Team 63: An Explainable Approach to Predicting NBA Game Winners

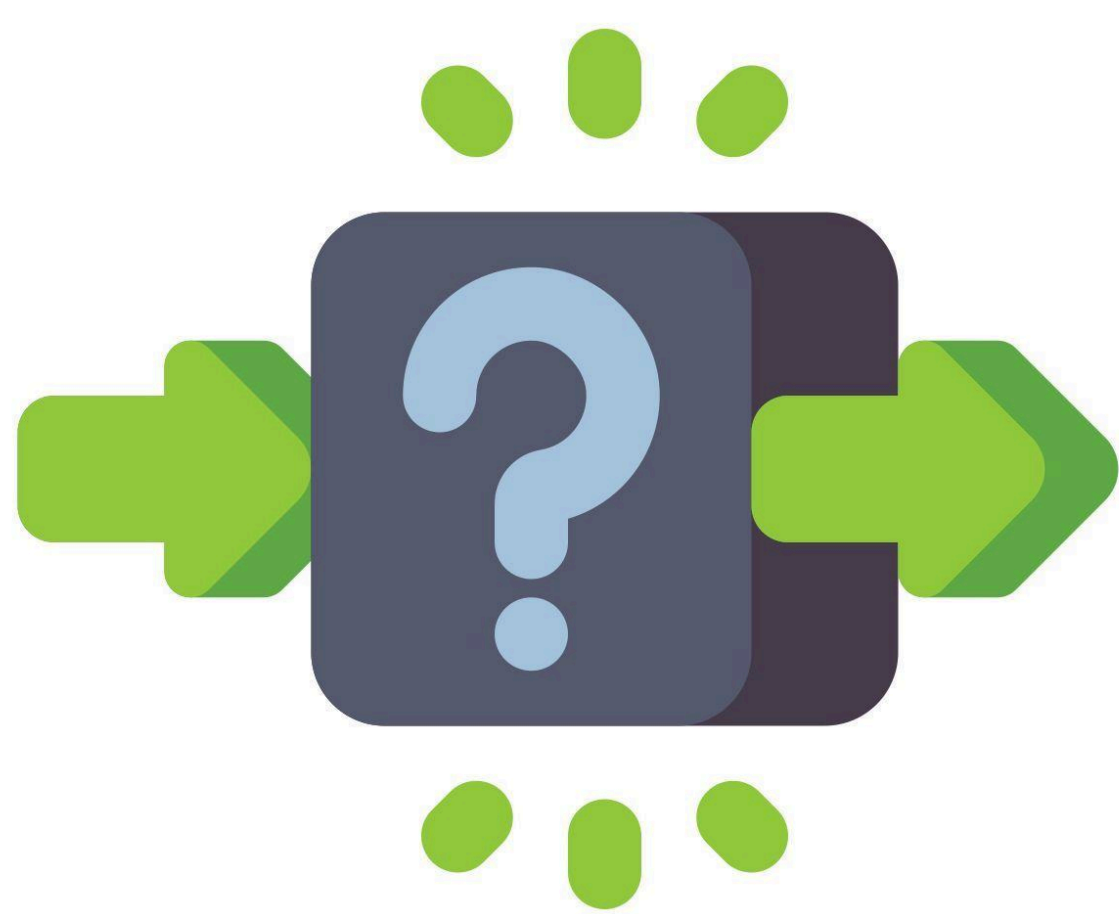
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Introduction

Predicting NBA game outcomes has traditionally relied on black-box models.

This project develops an explainable machine learning model for winner prediction, combining accuracy with transparent reasoning.

Each prediction is paired with a visual explanation to illustrate how specific factors influence the outcome.



Literature Survey

Walsh and Joshi [1], and Ren and Susnjak [11] demonstrate that calibration and probabilistic consistency are often more informative than raw model's accuracy.



Studies on game context further guide our feature engineering. Home-court advantage has been well-documented across leagues [2, 6], while fatigue and back-to-back scheduling effects are shown to reduce win probability in Esteves [3].

Evaluation

Do category scores (e.g. Endurance) match real game intuition?

We compared category scores with known team tendencies. Observations: High-efficiency teams scored higher in Offense.

Does the interface make explanations easy to understand?

We interacted with the radar chart, bar chart, and raw stats table. Observations: The radar chart made strength differences obvious at a glance.

Problem definition

Given a set of NBA games with features and final results, our goal is to build a model that delivers accurate yet interpretable predictions.



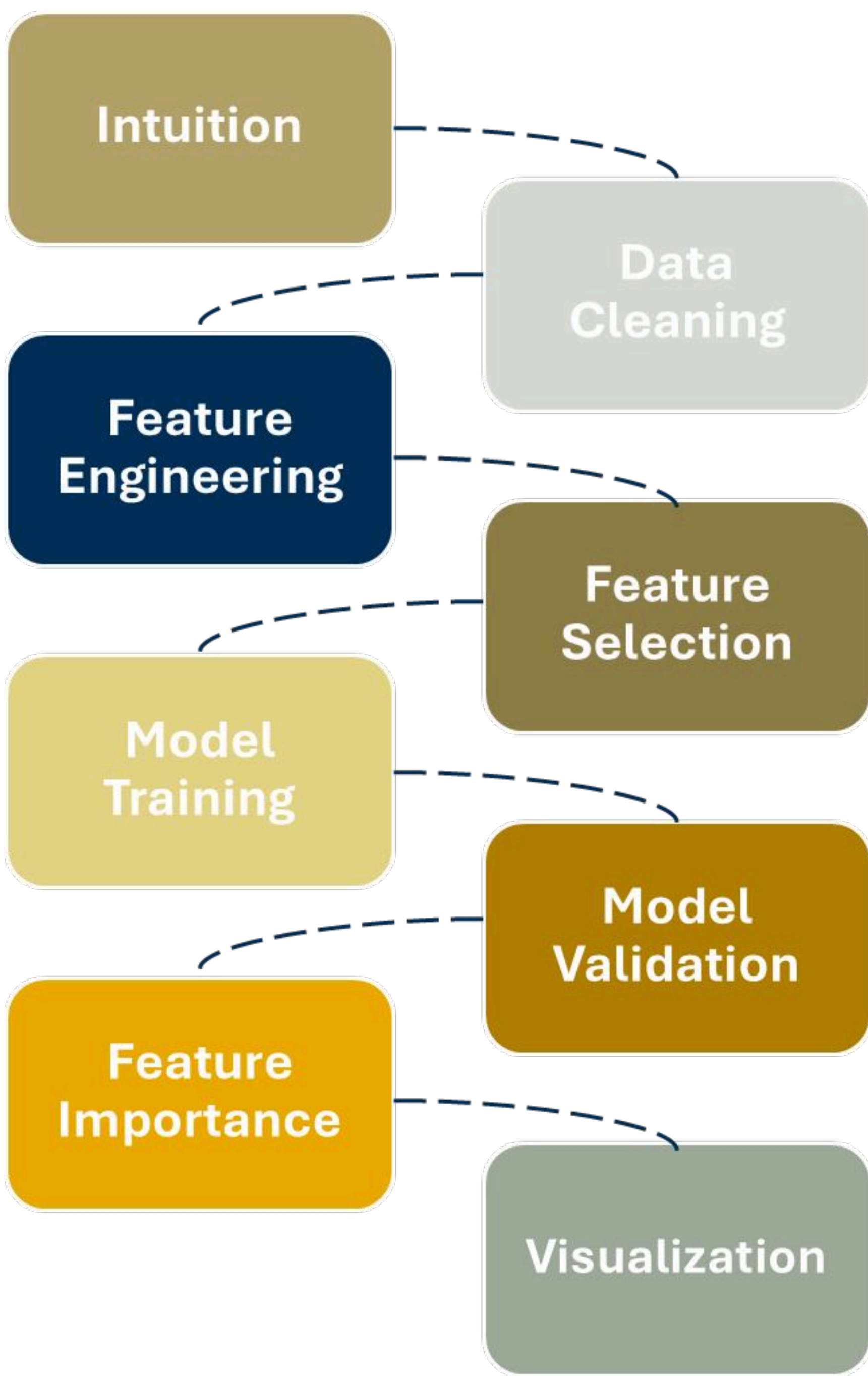
Historical data is stored in SQLite and new engineered features are created in Python, such as rest days, and recent performance. The model will be evaluated using accuracy, ROC-AUC, and F1 score.



Success is measured by the clarity of explanations provided, these final visualizations are created using D3.



Proposed Method



Conclusions and Discussion

New rolling window features played a significant role. Also, by grouping features into intuitive categories (e.g. Offense), users can clearly understand why a team is favoured.

Although the model can not match the accuracy of professional betting algorithms, it consistently produced easy to interpret predictions.

There are several directions for future work: incorporating injury reports, lineup changes, or real-time data could improve prediction quality.

Overall, this project provided meaningful insights and served as a useful educational tool for understanding data-driven sports analytics.

